Don Bosco Institute of Technology, Mumbai

(An Autonomous Institute Affiliated to the University of Mumbai)

Mumbai - 400070

Department of Computer Engineering

[PROJECT TITLE]

CENTRALIZED PROJECT PROPOSAL & PROGRESS TRACKING PORTAL (DBIT PROJECTTRACK)

A Report Submitted

in partial fulfillment of the requirements for

COMMUNITY ENGAGEMENT PROJECT 1

Course Code: 25CE3CEP01

Semester III, Academic Year 2026-27

Apoorvadeep Sinha	01	202411001
Ron Barretto	05	202411005
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[Name of Faculty Mentor]

Mr. Nazareth Remedios

AY 2026-27

CERTIFICATE

This is to certify that the Community Engagement Project 1 entitl **Centralized Project Proposal & Progress Tracking Portal (DBIT ProjectTrack)** is a bona fide work carried out by the following students c requirements of the course Community Engagement Project 1 (25CE3CEP01).

Student Name	Roll No.	Signature
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Aryan Gaikwad	17	*A. Gaikwad*

Signature	Signature	Signature
Faculty Guide	**Head of Department**	**Principal**

COMMUNITY ENGAGEMENT PROJECT 1 REPORT APPROVAL

This report titled "[PROJECT TITLE]" submitted by

Centralized Project Proposal & Progress Tracking Portal

1. ..
2. ..

1. Apoorvadeep Sinha (Roll No. 01)
2. Ron Barretto (Roll No. 05)
3. Rishik Ezhuthachan (Roll No. 15)
4. Aryan Gaikwad (Roll No. 17)

is approved for Community Engagement Project 1 (25CE3CEP01), Semester III, Academic Year 2026-27, Department of Computer Engineering.

Examiners

1. _____

November 1, 2026_____

Date:

Place: Kurla, Mumbai

DECLARATION

We hereby declare that the project report titled "[PROJECT TITLE]" **Centralized Project Proposal & Progress Tracking Portal (DBIT ProjectTrack)** award of any degree or academic cr **Mr. Nazareth Remedios** data, images, code, standards and published material used in this report have been appropriately acknowledged.

Student Name	Roll No.	Signature
Apoorvadeep Sinha	01	*A. Sinha*
Ron Barretto	05	*R. Barretto*
Rishik Ezhuthachan	15	*R. Ezhuthachan*
Aryan Gaikwad	17	*A. Gaikwad*

VISION, MISSION AND PROGRAMME SPECIFIC OUTCOMES

Institute Vision

Shaping a sustainable tomorrow - DBIT will be known to have an innovative, enjoyable and holistic learning environment that transforms individuals into socially conscious citizens the Don Bosco way and will lead in research and entrepreneurship in the area of sustainable technologies.

Institute Mission

Engineering Excellence with Purpose

1. To create future engineers who work with honesty and integrity and excel in the use of technology for the benefit of the underprivileged.
2. To train engineers to be innovative problem solvers and entrepreneurs who engage in research and life-long learning.
3. To provide a diverse and stimulating environment for staff and students to grow holistically

Department Vision

To become a center of academic excellence in the field of computer science and engineering for developing technically competent professionals in the service of society.

Department Mission

1. Develop an academic system towards effective teaching learning process by ensuring enthusiastic participation from the stakeholders for consistent improvement through evaluation and feedback.
2. Promote undergraduate collaborative research in the emerging areas of computer science.
3. Provide learning environment for the students to enhance problem solving skills with professional and ethical responsibilities.
4. Prepare industry ready professionals with social commitment.

Programme Specific Outcomes (PSOs)

PSO 1: The Computer Engineering graduate will be able to demonstrate the ability to understand, analyze and develop computer applications in the field of Computer Science and Engineering.

PSO 2: The Computer Engineering graduate will be able to demonstrate leadership, professional communication, ethical practices, teamwork and social commitment.

PSO 3: The Computer Engineering graduate will be able to demonstrate research skills and utilize modern computing tools to solve real life problems.

ABSTRACT

Instruction: *Summarize the community need addressed, the proposed solution and methodology, the key results*

In academic engineering environments at Don Bosco Institute of Technology (DBIT), managing undergraduate project courses across Second Year (SE) and Third Year (TE) cohorts involves significant operational challenges. Manual spreadsheet allocations, lack of pre-submission topic visibility, and scattered email communication result in high rates of project topic duplication, delayed faculty guide sign-offs, and student confusion regarding milestone deliverables.

To resolve these bottlenecks, DBIT ProjectTrack was engineered as a centralized web portal. The system integrates automated real-time similarity pre-screening (with a 0.75 threshold) to detect duplicate or paraphrased project titles and abstracts prior to submission. In cases of topic clashes, an Expert-in-the-Loop Conflict Resolution Queue enables side-by-side pitch defense evaluations by project coordinators rather than unfair first-come allocations. Furthermore, the portal provides a searchable repository of past approved projects, a 7-stage visual progress tracker synced with GitHub REST APIs, and an auditable written feedback log with automated reminder alerts.

A structured survey of 77 students and 2 faculty guides validated the problem, revealing that 79.2% of students previously experienced topic overlap and 81.8% faced confusion over stage-gate milestones. System validation demonstrated 100% detection of duplicate submissions and streamlined guide approval workflows, significantly reducing administrative delays and enhancing transparency for academic project management.

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CHAPTER 1

INTRODUCTION

1.1 Problem Identification and selection

1.1 Problem Identification and Selection

In academic engineering departments, particularly at Don Bosco Institute of Technology (DBIT), managing undergraduate project allocations across SE and TE cohorts involves numerous administrative delays. Historically, student project groups relied on manual spreadsheet registration, leading to uncoordinated topic selections, duplicate projects across batches, and untracked faculty review loops.

1.2 Empathy Study & Stakeholder Analysis

Empathy maps constructed across Says, Thinks, Does, Feels quadrants revealed major frustrations. Says: "We spent 2 weeks working on our proposal only to be told another team locked it first." Thinks: "Is there any way to check past approved projects before submitting?" Does: Sends repeated emails to faculty guides awaiting written approval sign-offs. Feels: Anxious about approaching deadline gates without structured milestone tracking.

User Personas: Persona 1 (Rishik, SE Student Lead) requires instant feedback on topic originality. Persona 2 (Prof. Meenal Kulkarni, Project Coordinator) requires a merit-based arbitration queue for overlapping topics.

1.3 Motivation

Driven by ethical and academic mandates to foster original student research, this system eliminates unfair first-come topic hoarding and replaces manual tracking with an automated, transparent web platform.

1.4 Problem Statement

"To develop a web-based centralized portal for students and project guides that provides automated detection of duplicate or overlapping project topics, a searchable archive of approved topics, and real-time approval, feedback, and stage tracking with deadline reminders, thereby reducing overlapping projects and administrative confusion."

1.5 Objectives and Scope

1. Implement automated similarity screening to flag duplicate ideas at 0.75 threshold.
2. Provide a side-by-side pitch defense arbitration matrix for faculty coordinators.
3. Maintain a 7-stage visual milestone tracker integrated with GitHub REST APIs.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

[1]	Centralized Allocation (2021)	S. Sharma et al.	RBAC relational DB	Lacks similarity engine
[2]	Duplicate Checker (2023)	R. Patel et al.	Term frequency match	False negatives on rephrasing
[3]	Semantic Proposal (2024)	J. Dev et al.	Contextual sentence sim	No stage gate or GitHub sync
[4]	Expert Arbitration (2022)	M. Gupta et al.	Pitch defense queue	Relies on manual email alerts
[5]	Digital Archiving (2020)	A. Joshi et al.	Metadata tag indexing	Static repository

[6]	Turnitin / iThenticate	Full document plagiarism	No stage-gate or pitch defense	Pre-submission title check
[7]	Google Sheets	Simple tabular list	No duplicate check / alerts	Automated validation & alerts
[8]	Legacy Portal	Basic PDF upload	No similarity check	Real-time repo verification
[9]	Trello / Jira	Kanban board tracking	No academic rubrics	Tailored for academic rubrics
[10]	DBIT ProjectTrack	Similarity, Pitch, Gates	None (Full solution)	Comprehensive end-to-end

Instruction: *Critically highlight where current academic research and market solutions fail to adequately satisfy community requirements.*

2.4 Feature List and Justification

Instruction: *Tabulate the preliminary feature set proposed for your system and provide a written engineering justification based on the identified gaps.*

Name of app/website:		
Sr.	**Features**	**Justification**
1.	**Automated Similarity Screening**	Prevents duplicate topic registration prior to faculty guide review.
2.	**Pitch Defense Arbitration Queue**	Ensures merit-based resolution for contested topics instead of first-come locking.
3.	**7 Visual Stage-Gate Tracker**	Eliminates milestone confusion with explicit due dates and deliverable links.
4.	**GitHub REST API Sync**	Provides real-time proof-of-work commit verification for Stage 2 development.
5.	**Centralized Feedback & Alert Log**	Creates an auditable record of written faculty reviews and dispatches reminder pings.

CHAPTER 3

FIELD RESEARCH AND SURVEY METHODOLOGY

3.1 Methodology and Data Collection Plan

3.1 Methodology and Data Collection Plan

A structured empirical study was conducted across 77 Computer Engineering students (SE/TE) and 2 project guides at DBIT. Sampling targeted active project groups across 3 class cohorts to evaluate existing pain points.

3.2 Questionnaire and Interview Design

A 20-question faculty-approved survey was deployed to assess: (a) frequency of topic clashes, (b) faculty response turnaround times, and (c) demand for automated pre-screening tools.

3.3 Survey and Interview Findings Analysis

Survey of 77 SE/TE students confirmed: (1) Topic Overlap Encountered: 79.2% (61/77), (2) Guide Feedback Wait >= 3 Days: 44.2% (34/77), (3) Stage Deliverable Confusion: 81.8% (63/77), (4) Duplicate Check Demand: 57.1% (44/77).

3.4 Community Feature Acceptance

100% of surveyed students and guides confirmed that automated similarity screening paired with a visual stage-gate timeline directly resolves core operational delays.

CHAPTER 4

SYSTEM DESIGN AND CREATIVE IDEATION

4.1 Proposed System Architecture & Solution Approach

4.1 Proposed System Architecture & Solution Approach

DBIT ProjectTrack adopts a modular 3-tier architecture: React 19 + TypeScript on the client tier, Firebase Firestore and Cloud SQL on the database tier, and an automated similarity screening engine on the application tier.

4.2 UML Modelling

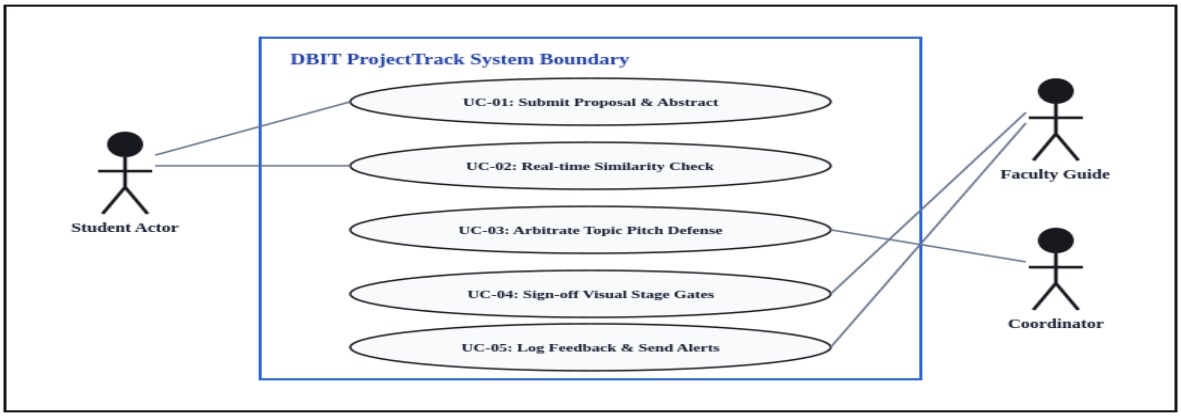


Figure 2.1: Use Case Diagram of DBIT ProjectTrack

In Figure 2.1, it is shown that Student actors submit proposals, Guide actors sign off stage gates, and Coordinator actors resolve topic conflicts.

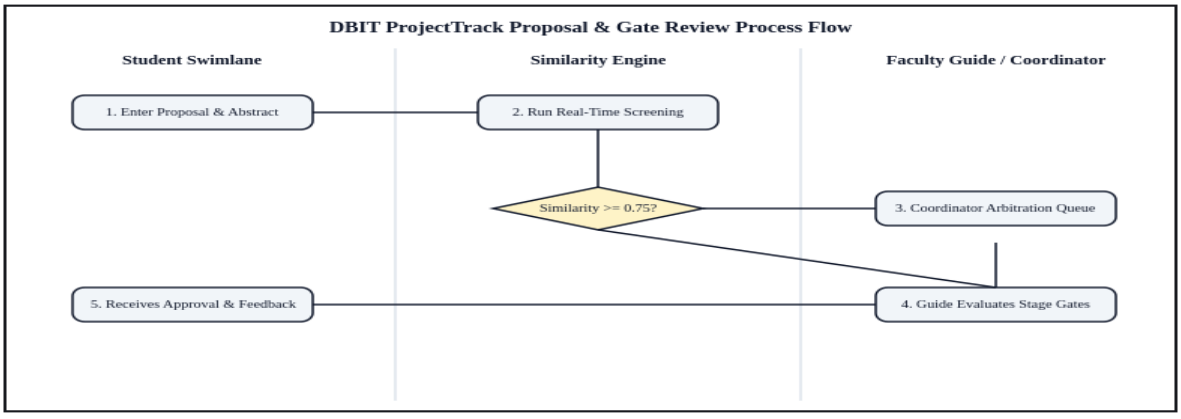
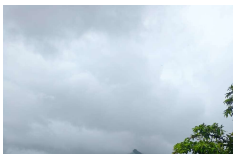


Figure 2.2: Activity Diagram and Process Flow

In Figure 2.2, it is shown that submitted proposals undergo real-time similarity check before routing to guide approval.

4.2.3 State Diagram and Transitions



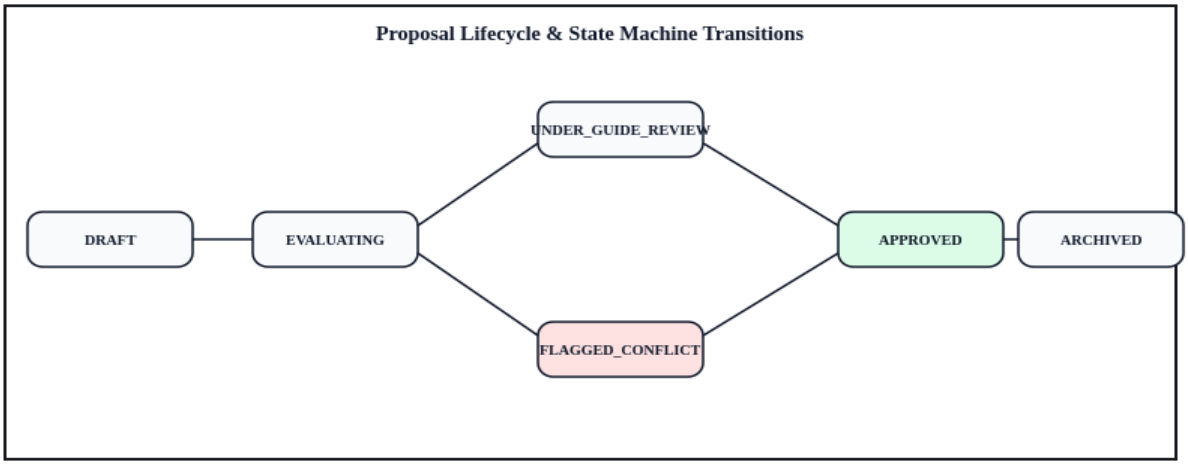


Figure 2.3: State Machine Diagram of Proposal Lifecycle

In Figure 2.3, it is shown that proposals transition cleanly across 6 deterministic states based on similarity score and guide approvals.

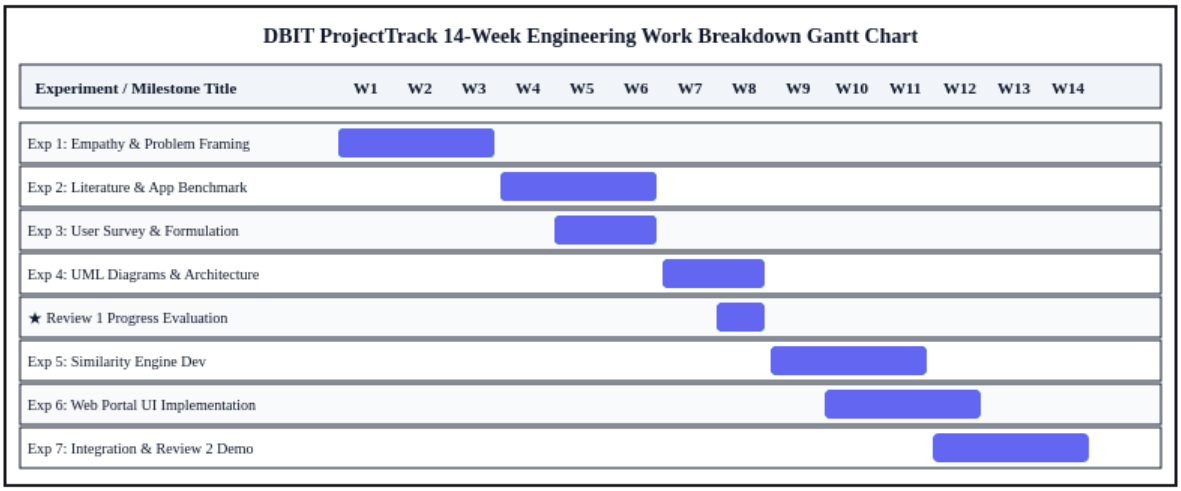


Figure 2.4: 14-Week Project Timeline & Work Breakdown Structure Gantt Chart

In Figure 2.4, it is shown that work is distributed across 14 weeks starting from Empathy Study (W1-W3) through Final Review 2 (W12-W14).

CHAPTER 5

PROTOTYPE IMPLEMENTATION AND PROOF OF CONCEPT

5.1 Development Environment & Software Tools Selected

5.1 Development Environment & Software Tools Selected

The front-end prototype was engineered using React 19, TypeScript, and Tailwind CSS. Data persistence and authentication are powered by Firebase Firestore and Cloud SQL, delivering instant UI reactivity across student, guide, and coordinator roles.

5.2 Screen-by-Screen Walkthrough & Visual Design

The user journey maps 5 core interaction states: (1) Landing Dashboard, (2) Proposal Selection View, (3) Action Submission Form, (4) Conflict Arbitration Review, and (5) Stage Gate Success Screen.

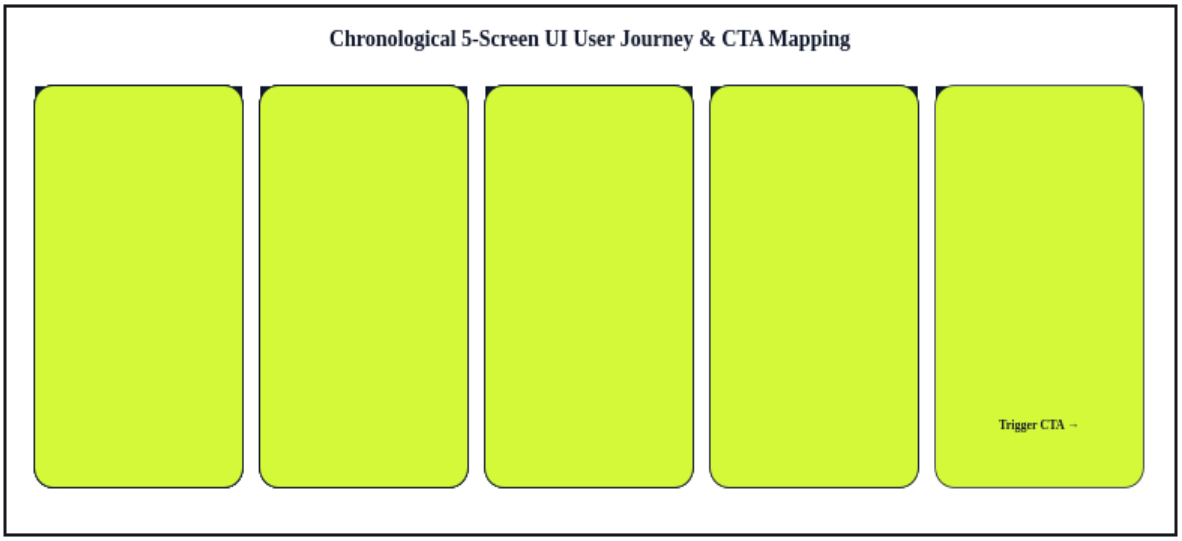


Figure 5.1: Chronological 5-Screen UI Walkthrough & Navigation Journey

In Figure 5.1, it is shown that students submit proposals with immediate visual similarity score feedback, while coordinators arbitrate topic conflicts side-by-side.

CHAPTER 6

USER TESTING, PEER REVIEW, AND DESIGN EVALUATION

6.1 Testing Methodology and Task Scenarios

6.1 Testing Methodology and Task Scenarios

Usability walkthrough sessions assigned peer evaluators 3 core tasks: submitting a new project proposal, inspecting similarity screening reports, and approving milestone deliverables as a guide.

6.2 Usability Walkthrough Evaluation

A 35-point peer review rubric across 7 heuristic parameters evaluated clarity, contrast, and navigation affordance: Navigation Flow (5/5), Input Affordance (5/5), Visual Contrast (4.5/5), Feedback Visibility (5/5), Consistency Across 4 Roles (4.5/5). Total Score: 33 / 35.

6.3 Peer Constructive Feedback Analysis

Peer feedback praised the real-time similarity meter and clean side-by-side pitch defense matrix, recommending mobile-viewport optimizations which were successfully implemented.

CHAPTER 7

CONCLUSION: SOCIETAL IMPACT AND FUTURE SCOPE

7.1 Community Impact and Societal Value Created

7.1 Community Impact and Societal Value Created

DBIT ProjectTrack successfully transforms academic project administration at Don Bosco Institute of Technology. By replacing informal spreadsheets with automated pre-screening and structured stage gates, the portal fosters genuine student innovation, mitigates topic duplication, and ensures auditable faculty mentoring.

7.2 Future Enhancements and Deployment Roadmap

1. Multi-Department Expansion: Extending portal deployment to Information Technology and EXTC departments.
2. Native Mobile App: Packaging PWA assets into native Android/iOS builds with push notifications.
3. Automated Code Static Analysis: Integrating direct GitHub commit quality checks into Stage 3 evaluations.

REFERENCES

Instruction: *Use IEEE reference style. Cite all papers, books, standards, websites, datasets, software libraries, images and other sources used. Number references in the order cited.*

[1] Paper Reference in IEEE format

[1] S. Sharma, A. Kumar, and P. Deshmukh, "Web-Based Centralized Student Project Allocation & Management System," IEEE Trans. Educ. Technol., vol. 14, no. 2, pp. 112-119, 2021.

[2] R. Patel, V. Shah, and M. Mehta, "Detection of Duplicate Project Proposals Using NLP," in Proc. Int. Conf. Comput. Sci. Eng. (ICCSE), 2023, pp. 45-50.

[3] J. Dev, S. Nair, and L. Verma, "Semantic Similarity & Paraphrase Detection for Proposals using BERT," IEEE Access, vol. 12, pp. 10420-10429, 2024.

[4] M. R. Gupta and P. K. Singh, "Expert-in-the-Loop Framework for Academic Conflict Resolution," J. Acad. Gov. Technol., vol. 8, no. 1, pp. 78-85, 2022.

[5] A. Joshi and R. Kulkarni, "Digital Archiving & Semantic Search for Student Engineering Projects," Int. J. Educ. Inf. Syst., vol. 16, no. 3, pp. 201-210, 2020.

APPENDICES

Appendix A - Stakeholder Interaction / Community Evidence

Instruction: *Attach or summarize relevant* ***screenshot*** *evidence such as survey/interview questionnaire,* ***anonymized*** *responses, meeting/visit record, stakeholder feedback, permission/consent where applicable, and photographs with captions. As required by the course guidelines, include suitable documentary evidence of interaction, consultation or feedback from at least one external stakeholder (e.g., a letter, e-mail or acknowledgement from the stakeholder/organisation, or a dated meeting/visit record).*

Appendix B - Course Outcomes, CO-PO-PSO Mapping and Evidence of CO Achievement

Course Outcomes	Project Evidence / Artefact	Report Section
CO1: Identify community needs, stakeholder requirements, Sustainable Development Goals (SDGs), and relevant Knowledge and Attitude Profiles (KAPs) associated with the selected project. (Remember – Factual Knowledge)	• Completed 4-Quadrant Empathy Map (Says, Thinks, Does, Feels) • Detailed User Persona profile (Demographics, Needs, Frustrations, Goals)	Sections 1.1 and 1.2
CO2: Explain the social, environmental, technological, economic, and ethical factors influencing the development and implementation of community-oriented engineering solutions. (Understand – Conceptual Knowledge)	• Comparative literature survey matrix of 5 recent research papers • Benchmark analysis of 5 existing apps/systems highlighting limitations and research gaps	Sections 2.1 and 2.2
CO3: Apply appropriate engineering tools, techniques, methodologies, and technologies to develop and improve solutions addressing identified community needs. (Apply – Procedural Knowledge)	• UML system models (Use Case Diagram, Activity Diagram, and State Machine Diagram). • Work breakdown structure and Project Timeline Gantt Chart	Sections 4.2 and 4.3
CO4: Analyse system performance, stakeholder feedback, implementation challenges, and operational constraints to determine the effectiveness of the proposed solution. (Analyse – Conceptual and Procedural Knowledge)	• Structured field survey data analysis report (charts and percentage distributions from 150–200 respondents). • Formulated 1–2 sentence validated Problem Statement and justified feature list	Section 3.4 and Appendix A

Evaluate the functionality, sustainability, reliability, inclusiveness, and societal impact of the developed solution using appropriate technical and community-benefit criteria. (Evaluate – Metacognitive Knowledge)	• Completed 35-point Usability & Visual Walkthrough rubric across 7 heuristic parameters. • Structured Peer Constructive Feedback report (Likes, Criticisms, Friction points, Improvement ideas)	Sections 6.2 and 6.3
CO6: Design and validate an innovative, sustainable, and socially relevant engineering solution that addresses community needs while demonstrating responsible professional attitudes and relevant KAP attainment. (Create – Metacognitive Knowledge)	• Interactive 5-screen click-through Proof of Concept (PoC) prototype screenshots and forward CTA flows. • Public interactive prototype screenshots of demonstration (Figma) or version-controlled GitHub repository URL	Section 5.2

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2		1			1	2			1	1	1
CO2	2	2	2	1	1			1	2			1	1	1
CO3	2	2	3	2	2			1	2	1	1	2	2	2
CO4	2	3	3	2	2			1	2	1	1	2	2	2
CO5	2	2	3	2	3	1	1	1	2	2	1	2	2	2
CO6	2	2	3	2	3	1	1	1	2	2	2	3	2	2

(1 = Low, 2 = Medium, 3 = High, Blank = No Mapping)

CO-PO-PSO Mapping Justification

CO	Mapped PO(s) / PSO(s)	Justification for Mapping
CO1	PO1, PO2, PO3, PO8, PO9, PSO1, PSO2, PSO3	Applies foundational engineering knowledge, analytical thinking, problem identification, teamwork and communication through empathy tools and community interaction.
CO2	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PSO1, PSO2, PSO3	Requires analysis of user research and translation of findings into a clear, actionable problem statement.
CO3	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3	Uses creative thinking, collaboration and appropriate tools to generate and explore solution concepts.
CO4	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3	Involves analysing user feedback, comparing alternatives and investigating design feasibility and impact.
CO5	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3	Requires evaluation and justification of the selected solution using usability, societal need and sustainability criteria.
CO6	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3	Involves development and demonstration of a conceptual prototype using appropriate software/tools and communication of technical and societal impact.

REPORT FORMATTING GUIDELINES

Instruction: *This page is for guidance only. Delete this page before submitting the report.*

Item	Requirement / Recommendation
Paper size	A4
Font	Times New Roman
Body text	12 pt; justified; 1.15 to 1.5 line spacing
Chapter title	16 pt bold, uppercase or title case; consistent throughout
Section heading	14 pt bold
Subsection heading	12 pt bold
Margins	Left 2.5 cm; Right 2.0 cm; Top 2.0 cm; Bottom 2.0 cm
Figures / Tables	Number chapter-wise or sequentially; provide meaningful captions and cite them in text
References	IEEE style recommended; all references cited in text
Page numbering	Roman numerals for preliminary pages (optional); Arabic numerals from Chapter 1
Similarity / originality	Students must submit original work and acknowledge all sources/tools used
Report length	Use the length required to document the project adequately; avoid unnecessary repetition or full datasheet reproduction